MARYAM SABERI

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING

PENN STATE UNIVERSITY

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INTRODUCTION

I am a Ph.D. candidate in Electrical Engineering at Penn State University with research expertise in state estimation, LiDAR-inertial odometry, SLAM, adaptive filtering, and sensor fusion. In the Control and Autonomous Robotics Lab (CARL), I develop adaptive estimation methods to improve localization and estimation accuracy under unknown IMU disturbances, unknown and potentially nonzero mean measurement noise, and model uncertainty. My recent work integrates these methods into Point-LIO and investigates efficient and robust localization for mobile robotic systems using simulation-based ROS/ROS 2 workflows.

EDUCATION

2022 – Present
PENNSYLVANIA STATE UNIVERSITY
Ph.D. Candidate in Electrical Engineering, Control and Autonomous Robotics Lab (CARL).
Expected completion: Spring 2027.
Research: Adaptive Kalman filtering, nonlinear estimation, robust LiDAR-inertial odometry, and compensation for unknown measurement and process disturbances.

2022 – 2024
PENNSYLVANIA STATE UNIVERSITY
Master of Science in Electrical Engineering, Control and Autonomous Robotics Lab (CARL).
Project: Adaptive Kalman Filtering for Systems Under the Influence of Unknown External Observation Noises.

2020 – 2021
UNIVERSITY OF ALABAMA AT BIRMINGHAM
Doctoral studies, Bio-inspired Integrated Circuit (BIC) Design Laboratory.
Research: Low-cost printed sensing systems and energy-efficient integrated circuits.

2013 – 2016
ISLAMIC AZAD UNIVERSITY OF MASHHAD
Master of Science in Biomedical Engineering – Bioelectric, Neuromuscular and Rehabilitation Lab.
Thesis: Adaptive muscle synergy modeling for elbow muscle activation during hand-reaching.

2008 – 2012
SADJAD UNIVERSITY OF TECHNOLOGY
Bachelor of Science in Electrical Engineering.
Thesis: Design and fabrication of an electrical nursing system for vital-sign monitoring.

RESEARCH INTERESTS

Robotic localization, LiDAR-inertial odometry, SLAM, sensor fusion, robust and adaptive state estimation, unknown sensor disturbance compensation, point cloud mapping, autonomous robotics, nonlinear filtering, control theory.

PROFESSIONAL EXPERIENCE

Fall 2022 – Present Research Assistant, Control and Autonomous Robotics Lab (CARL), Penn State

- **Point-LIO and robust LiDAR-inertial odometry:** Investigated the impact of unknown external IMU disturbances on Point-LIO localization and developed an adaptive EKF based disturbance compensation approach to improved robustness.
- **Efficient state estimation:** Developed a reduced-state Point-LIO formulation that decreases the estimator state dimension from 24 to 18 (25% reduction), while incorporating online estimation of unknown sensor disturbances.
- **Adaptive estimation methods:** Developed and evaluated adaptive Kalman filtering algorithms for unknown measurement noise, nonlinear observation errors, and simultaneous measurement/model uncertainty.
- **Validation and dissemination:** Conducted numerical simulations, Monte Carlo evaluations, error comparisons, and computational benchmarking.

Spring 2024 – Present Teaching Assistant, Circuits and Devices (EE 210), Penn State
Supported laboratory experiments, circuit debugging, and instruction in fundamental electrical engineering measurements and devices.

2020 – 2021 Research Assistant, Bio-inspired Integrated Circuit Design Laboratory, UAB
Researched sensing and classification methods and low-cost printed sensing technologies.

2015 – 2017 Technical Manager, Avin Exon Pars Ltd

SOFTWARE SKILLS

Robotics and simulation: ROS, ROS 2, Point-LIO, Fast-LIO, NVIDIA Isaac Sim; LiDAR/IMU data processing and robotics simulation.
Estimation and analysis: Kalman filter, EKF, UKF, adaptive filtering, nonlinear state estimation, sensor fusion, Monte Carlo evaluation.
Programming and computing: MATLAB, Python, Linux.
Additional tools: Multisim, LaTeX, Microsoft Word, Excel, PowerPoint.

PUBLICATION

M. Saberi, Hung-Chieh Wu, and D. Ebeigbe, "Adaptive LiDAR-Inertial Odometry with Online Estimation of Unknown IMU Disturbances," manuscript submitted to *IEEE Robotics and Automation Letters*, under review.

M. Saberi and D. Ebeigbe, "Adaptive Kalman Filter for Simultaneous Compensation of Unknown External Measurement Noise and Model Uncertainty," manuscript submitted to *IEEE Control Systems Letters*, under review.

Hung-Chieh Wu, **M. Saberi**, Kasra Sinaei, and D. Ebeigbe, "PRECONET: A Kalman-Filter-Inspired Predictor–Corrector State Estimator via Massively Parallel Reinforcement Learning," manuscript submitted to *IEEE Robotics and Automation Letters*, under review.

M. Saberi and D. Ebeigbe, "Reducing Observation Linearization Errors of the Extended Kalman Filter via Adaptation," *2026 American Control Conference (ACC)*, New Orleans, LA, USA, 2026, pp. 2805–2810.

M. Saberi and D. Ebeigbe, "Adaptive Kalman Filtering for Unknown External Measurement Noise," *IEEE Signal Processing Letters*, vol. 33, pp. 868–872, 2026, doi: 10.1109/LSP.2026.3657705.

A. U. Alsaggaf, **M. Saberi**, T. Berry, and D. Ebeigbe, "Nonlinear Kalman Filtering in the Absence of Direct Functional Relationships Between Measurement and State," *IEEE Control Systems Letters*, vol. 8, pp. 2865–2870, 2024, doi: 10.1109/LCSYS.2024.3514818.

M. Saberi, S. D. Gardner, and M. R. Haider, "A Machine Learning Based Smart Contact-less pH Sensing and Classification," *2021 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, Lansing, MI, USA, 2021, pp. 1049–1052, doi: 10.1109/MWSCAS47672.2021.9531918.

AWARDS

Melvin P. Bloom Memorial Graduate Fellowship in Electrical Engineering (Spring 2024).